

# Homework 2

## *Sommerfeld Model*

1. (30 pts) In this problem we will compare and contrast the “particle in a box” to free electrons with Born-von-Karman Boundary conditions.
  - (a) Assume a 1-d box of size  $L$  with a single-particle. The potential is infinite for  $x \leq 0$  and  $x \geq L$ ; and otherwise zero. Solve for the Eigenvalues and Eigenfunctions; make note of the smallest Eigenvalue. Also, compute the density of states assuming that  $L$  is very large.
  - (b) Assuming free electrons and BvK boundary conditions (for system size  $L$ ) in 1-d, find the Eigenvalues and Eigenfunctions. Also, compute the density of states assuming that  $L$  is very large.
  - (c) Compare and contrast the smallest Eigenvalue for “particle in a box” vs free electrons with BvK boundary conditions. If this answer is different, why? What are the implications.
2. (30pts) Consider the following scenarios:
  - (a) Assuming free Fermions, estimate the Fermi temperature of liquid  $^3\text{He}$  (mass density  $\rho_m = 8.1 \times 10^{-2} \text{ g/cm}^3$ ).
  - (b) Assuming free Fermions, estimate the Fermi temperature of neutrons in a neutron star ( $\rho_m = 10^{14} \text{ g/cm}^3$ ).
  - (c) At room temperature the resistivity for Cu is  $\rho = 1.78 \times 10^{-6} \Omega\text{-cm}$ . What is the mean free path of the conducting free electrons? (the lattice parameter for Cu is  $3.61 \text{ \AA}$ ).
3. (35 pts) Consider a thin square layer of silver that is  $10^6 \text{ \AA}$  long in the plane defined by the  $x$  and  $y$  directions. Treat the layer as a free gas of Fermions for motion in the  $(x, y)$  plane. Electron motion in the  $z$ -direction normal to the plane is assumed to be an infinite barrier quantum well (ie. particle in the box) whose size is given by the thickness of the layer (defined below).
  - (a) Take the layer to be  $4.1 \text{ \AA}$  thick along the  $z$ -direction. Find the Eigenvalues of the system, and plot the DOS; and determine the Fermi energy.
  - (b) Repeat with a layer of  $8.2 \text{ \AA}$  thick along the  $z$ -direction.
4. (5 pts) In class we derived the energy as a function of temperature (low T limit) for Free Fermions using a Taylor series. Let us obtain a crude estimate by relying on the fact that only electrons within  $k_B T$  of the Fermi energy are excited, yielding  $g(\epsilon_F)k_B T$  electrons per volume being excited. Approximate the internal energy as a function of temperature, and obtain heat capacity. Compare to the more precise Taylor series result.